

ELYSE KIMIYE PISKACEK

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Education

University of British Columbia

Bachelor of Applied Science | Biomedical Engineering | Expected Graduation: May 2028

Vancouver, BC

Sept. 2024 – Present

Experience

UBC BEST

ENABLE Co-Lead

Mechanical Engineer

Vancouver, BC

August 2026 – Present

Sept 2025 – July 2026

- Lead a multidisciplinary team of ~15 students developing client-centered assistive technology solutions through **mechanical design** and **rapid prototyping**.
- Manage **project selection, recruitment, team coordination, and client outreach**, translating client needs into engineering requirements and project opportunities.
- Previously designed and prototyped solutions as a Mechanical Engineer, using **SolidWorks** and **3D printing**

Engineers Without Borders | ACCESS Venture

Mechanical Engineer

Vancouver, BC

Nov. 2025 - Present

- Designed and prototyped a low-cost urine collection device for female HPV self-collection testing in Sub-Saharan Africa.
- **3D printed** and **molded prototypes** in PLA and silicone, iterating the design based on water-based testing of the collection mechanism.

Projects

Neonatal Measuring Device | May 2026 – Sept. 2026

BC Children's Hospital

- Designed and prototyped a caliper-style infant measuring device, incorporating a telescoping mechanism, guarded measuring tape, and weight-stabilized headrest.
- **3D printed and assembled** a physical prototype, which was provided to a BC Children's Hospital researcher for use in an ongoing study comparing infant measurement methods.

Finger Physio Stretching Device | Sept. 2025 – July 2026

Technology for Living

- Designed a modular finger-stretching device with a custom hand platform modeled to the client's hand geometry.
- Developed **U-joint, worm-gear, and stepper motor mechanisms** for controlled finger movement and stretching.

Femoral Fracture Fixation Plate | Sept. 2025 – Nov. 2025

School of Biomedical Engineering

- Designed a minimally invasive **C-shaped double-plate fixation system** for elderly femoral fracture patients requiring immediate full weight bearing.
- Performed **finite element analysis** and selected as the top project by VGH trauma surgeon Dr. Pierre Guy.

Streamedical | July 2022

Shad Canada

- Developed an emergency department solution within a 25-day design challenge, focusing on business strategy and product development.
- Presented the final proposal and received "Best Business Proposal."

Community Engagement

Associate (Part-time) | May. 2026 – Present

PetSmart

- Manage customer transactions, inventory, and daily animal care while maintaining accurate records and store standards.

Activity Volunteer | Oct. 2022 – June 2024

Robert Nimi Nikkei Care Home

- Organized and led weekly art classes for resident seniors to promote creativity and engagement

Netball Coach | June 2023 – May 2024

BC Netball

- Introduced netball to 200+ elementary students and led practices and tournaments for groups of 30+ athletes.

Technical Skills

CAD & Design: SolidWorks, Onshape, Medical Device Design, Device Development

Mechanical: 3D Printing, Rapid Prototyping, Finite Element Analysis, Soldering

Programming: C, C++, C#, Java, Python, R, MATLAB, Simulink, Arduino IDE, GitHub

Instrumentation: IMUs, Strain Gauges, EMG, Wearable Sensors, Multimeter, Oscilloscope, Spectrophotometer